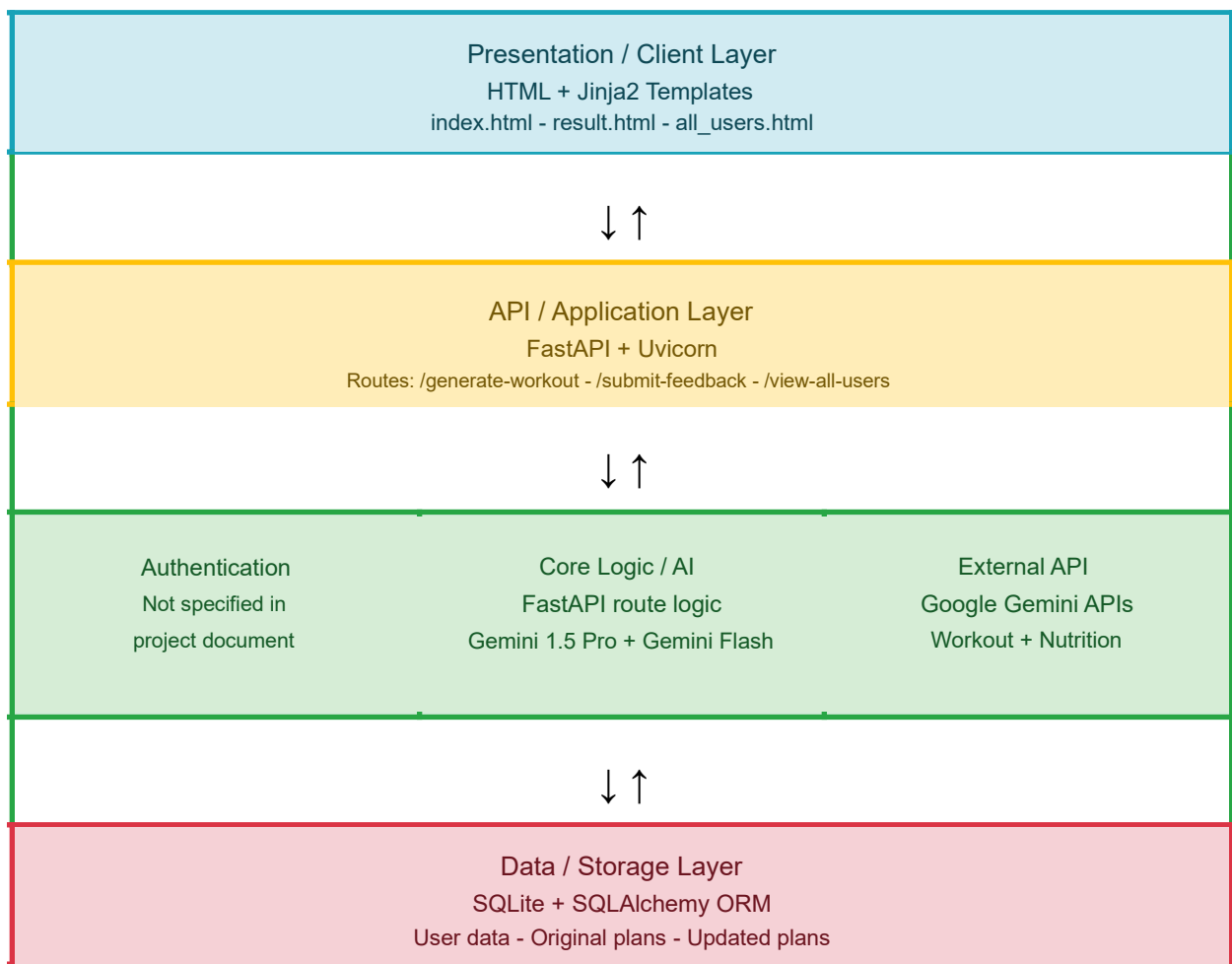


Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-6111
Project Name	FIT BUDDY AI
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer [Presentation Layer]	Collects user details, displays the generated 7-day plan and [Describe what this component does] nutrition tip, accepts feedback, and provides the admin view.	HTML, CSS, Jinja2 [List specific tech]
API / Application Layer [API Gateway]	Handles HTTP routes, form input, validation, Gemini calls, [Describe what this component does] plan updates, and dynamic template rendering.	FastAPI, Uvicorn, Pydantic [List specific tech]
Core Logic / AI Services [Microservices]	Generates workout plans, nutrition/recovery tips, and [Describe what this component does] feedback-based plan revisions using Gemini models.	Gemini 1.5 Pro, Gemini Flash, Python [List specific tech]
Database [Database]	Persists user information and AI-generated plans; stores [Describe what this component does] original and feedback-updated plans separately.	SQLite, SQLAlchemy ORM [List specific tech]